Superconducting Microresonator Signal Denoising Using Machine Learning

Viktor Makhrinov, Zhansaya Maksut, Adnan Yazici, *Senior Member, IEEE*, Akhan Almagambetov, Bruce Grossan, and Mehdi Shafiee

***Abstract*—Superconducting microresonators are pivotal in the realm of quantum computing, essential for establishing superpositions. As quantum technology evolves, these components become even more integral, not just in quantum applications but also as precision instruments in astronomy—specifically as highly sensitive cameras. Microwave kinetic inductance detectors are notable in this field, recognized for their heightened sensitivity crucial for both scientific exploration and industrial use. Yet, electronic noise remains a formidable obstacle. Addressing this, our approach combines machine learning strategies with traditional noise reduction techniques to enhance the signal clarity of superconducting microresonators. Our process involves gathering data and applying various denoising models, placing a significant emphasis on machine learning methods. We have discovered that certain denoising approaches, especially deep learning frameworks such as long short-term memory networks and autoencoders, are highly effective in clarifying the signals from superconducting microresonators. These models demonstrate versatility in handling different noise types, skill in deciphering intricate noise structures, and they consistently improve their performance as more data is provided.**

***Index Terms*—Denoising, electronic noise, signal reliability, superconducting microresonators.**

I. Introduction

SUPERCONDUCTING microresonators are essential in both quantum computing and imaging applications, particularly when integrated with qubits or microwave kinetic inductance detectors (MKIDs). Within the realm of qubits, these microresonators are indispensable tools for modifying and controlling quantum states. Their integration with qubits facilitates coherent interactions that are fundamental for executing quantum operations such as entanglement and superposition [1]. In the use of MKIDs, superconducting microresonators are critical for detecting and processing electromagnetic signals with exceptional sensitivity. Their capability to manage microwave signals renders them invaluable in fields such as quantum information processing and astrophysics research [2]. In our specific design, interdigitated capacitors are employed to finely tune resonant frequencies, while changes in inductance are facilitated through meandered lines [3]. Superconducting microresonators are affected by three primary noise types. Two-level system noise arises due to electrical variations between two energy states on the substrate surface [4]. Generation-recombination noise results from changes in the number of quasiparticles [5]. Lastly, readout noise originates from the cryo-amplifier or during the signal capture and conversion process [6]. These noise sources significantly challenge the effective operation of microresonators.

Recently, we have been investigating strategies to minimize two-level system noise and recombination noise by experimenting with different superconducting materials. Meanwhile, electronic noise typically lends itself to reduction through data processing techniques. To reduce electronic noise in superconducting microresonators, we have explored machine learning approaches for signal denoising and compared their performance with traditional methods such as wavelet denoising and Savitzky–Golay filtering. Traditional methods, while effective under certain conditions, can become inadequate or impractical in specific scenarios. When dealing with large datasets containing many pixels, traditional methods often prove computationally intensive and time-consuming, making data reduction inefficient. In cases involving rapid signal fluctuations or transient phenomena, these methods may fail to capture critical details, resulting in a loss of essential information. In addition, for time-sensitive measurements exhibiting nonstationary behavior, traditional approaches can introduce delays, reducing responsiveness and accuracy.

Our machine learning algorithms tackle these challenges by facilitating real-time data processing and improving the signal-to-noise ratio (SNR). These techniques enhance both accuracy and efficiency, making them highly effective in handling complex and noisy environments. In comparison to circle fitting algorithms [7], which are effective for extracting resonator parameters under ideal conditions with clean data, real-world scenarios often involve noise that obscures signals and introduces significant uncertainties. Machine learning methods tackle these

Received 9 June 2024; revised 3 November 2024 and 24 November 2024; accepted 24 November 2024. Date of publication 9 December 2024; date of current version 26 December 2024. This work was supported in part by the Nazarbayev University, Astana, Kazakhstan, under Grant 20122022FD411, and in part by the Kazakhstan Ministry of Science under Grant AP19676308. *(Corresponding author: Mehdi Shafiee.)*

Viktor Makhrinov and Akhan Almagambetov are with the Department of Electrical and Computer Engineering, Nazarbayev University, Astana 010000, Kazakhstan.

Zhansaya Maksut and Adnan Yazici are with the Department of Computer Science, Nazarbayev University, Astana 010000, Kazakhstan.

Bruce Grossan is with Energetic Cosmos Laboratory, Nazarbayev University, Astana 010000, Kazakhstan, and also with the Space Sciences Laboratory, University of California, Berkeley, CA 94720 USA.

Mehdi Shafiee is with the Department of Electrical and Computer Engineering, Nazarbayev University, Astana 010000, Kazakhstan, and also with Energetic Cosmos Laboratory, Nazarbayev University, Astana 010000, Kazakhstan (e-mail: mehdi.shafiee@nu.edu.kz).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TASC.2024.3513769.

Digital Object Identifier 10.1109/TASC.2024.3513769

1051-8223 © 2024 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

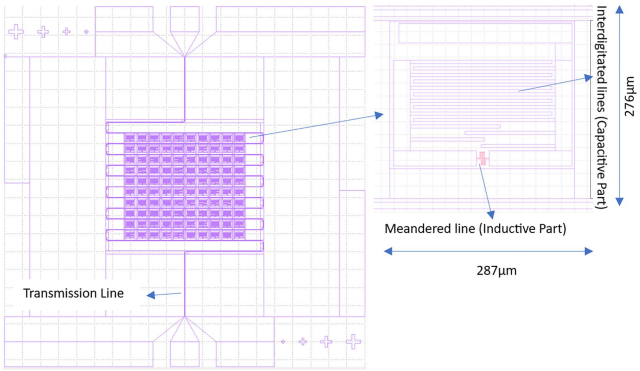

Fig. 1. Configuration of our 100-pixel superconducting microresonators includes interdigitated capacitors and meandered lines, all multiplexed within a single transmission line.

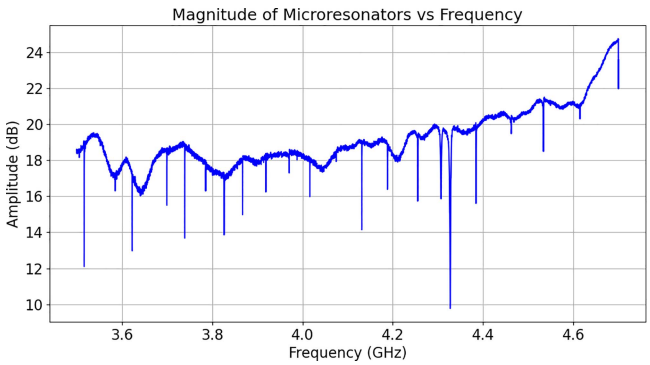

Fig. 2. Measured amplitude (in dB) of the S_{21} parameter from our microresonators.

challenges by simultaneously denoising the signal and enhancing the extraction of parameters such as resonant frequency and quality factor, thereby minimizing the risk of uncertainties in measurements.

In this article, our primary goal is to rigorously assess the effectiveness of these denoising techniques in improving the reliability and accuracy of microresonator signal analysis.

A. *Superconducting Microresonators and Experimental Setup*

In our research, we utilized superconducting microresonators constructed with a three-layer composition of TiN/Ti/TiN (3/10/3 nm), each shorted by 100 nm Al across all areas except the inductive segment. These devices operate at a critical temperature of 1.1 K and are built on a high resistivity silicon substrate. The superconducting microresonators designed for this study comprise 100 pixels, with resonant frequencies spanning from 3.5 to 7.2 GHz. Fig. 1 presents the design of our 100-pixel microresonator arrays, highlighting a single pixel configuration [8], [9].

Fig. 2 presents a selection of 20 pixels from the total 100 pixels in the magnitude measurements of our microresonators.

To analyze the performance of microresonators, we measure the S_{21} parameter, which is defined as 10*log (microresonators output power/microresonators input power). For our experiments, we supplied an input power of -90 dBm by using

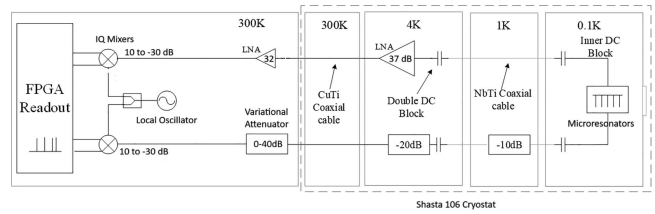

Fig. 3. Picture shows the electrical transmission line of our experimental setup, both inside and outside the cryostat, used for measuring microresonators.

attenuators and recorded the output power after amplification. All calculations were performed using the field programmable gate arrays (FPGAs) in our custom-designed readout system, which utilizes software-defined radio technology [10], [11].

During these measurements, the temperature of the microresonators was maintained at 100 mK within a cryostat (Model: SHASTA 106 from HPD Company), ensuring optimal conditions for our experiments. To minimize noise and improve the SNR, a HEMT amplifier with a 37 dB gain was used at the 4 K stage of the cryostat. Additional amplification was conducted outside the cryostat, at room temperature, using an amplifier with a 32 dB gain. To avoid saturation, the excitation input power for the microresonators is attenuated by three attenuators placed at different stages of the experimental setup inside and outside of cryostat. Fig. 3 illustrates our experimental setup, detailing the components and configuration used in our study.

Microresonators are constrained by various sources of noise, particularly from the amplifiers and readout systems. Despite improvements in electrical transmission lines aimed at reducing noise, some disturbances still affect the measurements. To tackle this challenge, we investigated several offline processing denoising techniques specifically for microresonators. These methods were evaluated for their impact on key parameters such as the microresonators' quality factor, mean square error (MSE), central frequency shift, and amplitude. Among the techniques examined, the merit method emerged as particularly effective. It not only preserved the central frequency and amplitude of the signals but also managed to correct the quality factor and smooth the flat portions of the signal, leading to a reduction in MSE.

II. Mathematical Algorithms for Denoising

We explore a diverse range of denoising techniques, each with unique characteristics tailored to address the challenges posed by microresonator signals.

There are some traditional methods of signal denoising, which can be applied to this particular task. For example, using mathematical techniques called wavelets, one can separate the original signal into its various frequency components and analyze each one separately [12]. Savitzky–Golay filter works well for smoothing signals, which raises the SNR without changing the signal itself [13]. Using initialization and propagation of error covariance statistics, the Kalman filter eliminates anomalies or disruptions from the signal [14], [15]. A nonlinear resilient approach called median filtering can be used to recover signals that are concealed in noisy data [16]. Another filtering method

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

that can be used to evaluate the signals is Wiener filtering [17], [18]. Considering both the noise and the desired signal to be stochastic processes with linear features, the Wiener filter filters noisy signals [19]. Another signal-denoising model that effectively preserves signal properties is the total variation (TV) regularization model [20]. Higher graph frequency values are attenuated while lower frequency items are retained by the low-pass filter [21]. As in [22], the goal of Gaussian filters is to maintain edge detail. Moving average smoothes the signal by taking the average of its nearby samples, suppressing extra peaks, and bringing it closer to the real shape without reducing the amplitude of the R-peak [23].

We were also keen to assess the efficacy of machine learning models in the denoising of microresonators. One of the most prominent and widely used deep learning networks is the convolutional neural network (CNN) [24], [25]. CNN is composed of layered convolutional layers that can automatically extract higher level features of input data using convolutional kernels [26]. During training, CNNs adjust to the noise distribution in input signals and learn to discriminate between noise and signal components. This resilience allows CNNs to successfully reduce noise while retaining signal properties, making them ideal for denoising microresonator signals. While they excel at capturing spatial relationships and complicated features, their adaptation to various noise characteristics is determined by the training data's diversity and representativeness [27].

The autoencoder (AE) is an unsupervised type of neural network structure that uses two training processes—encoding and decoding—to compress and rebuild data in order to extract features for the samples [28]. By learning compact representations of clean signal characteristics and eliminating noise during the reconstruction process, AEs show remarkable efficacy in denoising microresonator signals. Although they have a great degree of tolerance to various noise characteristics, the architectural design and hyperparameters used during training may have an impact on how well they operate.

The random forest regressor (RFR) is reasonably simple to develop, operates effectively on big datasets, retains accuracy on datasets with substantial data gaps, and provides a way to generate an uncertainty metric for the final prediction [29]. RFR is highly effective at denoising microresonator signals because it captures nonlinear linkages and interactions between input characteristics and output targets.

Recurrent neural networks (RNNs) are neural network models that process sequential data one element at a time and have the capacity to selectively convey information across sequence steps [30]. RNNs perform well in denoising microresonator signals by identifying short-term temporal dependencies and extracting significant features from noisy input sequences. While they are extremely adaptable to various noise characteristics, they may struggle to capture long-term dependencies in noisy input sequences, which Deep RNN can handle.

Deep recurrent networks are commonly designed using a particular type of building block known as long short-term memory (LSTM) units [31]. It accurately captures long-term dependencies and keeps relevant information in noisy input sequences. Their performance may vary depending on the architecture design and hyperparameters selected during training.

A. Denoising Models' Architecture

In this section, we delve into the architecture of the denoising models utilized for the 1000 noisy signals from microresonators. The architecture of the models has been meticulously crafted to accommodate the unique characteristics of the signals while effectively combating noise interference.

1) Convolutional Neural Network: The network architecture includes two 1-D convolutional layers, with the first layer containing 128 filters and the second 256 filters, both utilizing a kernel size of 5 and rectified linear unit (ReLU) activation $[\sigma(x) = \max(x, 0)]$. The network aims for arbitrary smoothness of the regression function, achieving near-minimax rates with ReLU activation in a deeper network structure [32]. Each convolutional layer is followed by an average pooling layer with a pool size of 2, which reduces the spatial dimensions and computational load. The outputs are flattened into a 1-D vector for transition to dense layers. Two dense layers follow, each containing 256 and 128 units, respectively, using ReLU activation. A final dense layer with a single unit is included to handle regression tasks. The model trains for 100 epochs with a batch size of 16. The optimization uses the adaptive moment estimation (Adam) optimizer, and performance is measured by the mean squared error (MSE) loss function. The Adam optimizer adjusts learning rates for various parameters based on gradient data [33]. This CNN configuration is fine-tuned for denoising and regression analysis, leveraging sequential layer processing and training parameters to predict continuous outcomes accurately.

2) Random Forest Regressor: RFR is implemented with 100 decision trees, enhancing the model's predictive accuracy and robustness through ensemble averaging. The model is trained on sequences of signal data that have been flattened and scaled to ensure uniformity in feature importance. Training utilizes random subsets of data for constructing each tree, improving the model's generalization capabilities across diverse datasets. This configuration of the RFR, with its emphasis on ensemble techniques and careful data preparation, is optimized for accurate regression predictions in complex data environments.

3) Sparse AE: The sparse AE is structured into two main components. The encoder layer comprises two dense neural network layers, starting with 128 units, followed by a smaller intermediate layer with 32 units, both employing leaky ReLU activation functions. The encoder aims to compress the input data into a lower dimensional representation, effectively capturing essential features while reducing data redundancy. The decoder, similar in structure to the encoder, features two dense layers, with the final layer having the same number of units as the original input data. It uses leaky ReLU activation to reconstruct the data from its encoded form, seeking to produce an output that closely approximates the original input. The network is trained for 300 epochs with a batch size of 64, enabling efficient learning without overloading computational resources. The Adam optimizer, enhanced with weight decay for regularization, is

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

used for its adaptive learning rate capabilities, aiding in faster convergence to the global minimum. A combined MSE and MAE loss function measures the reconstruction error, helping to minimize the difference between the original and decoded data. This improved sparse AE setup effectively learns to encode and decode input data, making it suitable for applications requiring data compression and feature extraction while retaining critical information and minimizing reconstruction errors.

4) Long Short-Term Memory: The network incorporates two LSTM layers, each consisting of 50 units. The first LSTM layer maintains temporal sequence information throughout the layers, which is crucial for capturing time-dependent patterns in the data. A dense layer with 50 units follows, retaining the temporal characteristics learned by the LSTM layers. A final single-unit dense layer is used for regression tasks, enabling the network to predict continuous output values. The model undergoes training over 50 epochs with a batch size of 32, balancing computational efficiency with training thoroughness. Training employs the Adam optimizer for its efficient adaptive learning rate adjustments, coupled with the MSE loss function to quantify and minimize prediction errors. This LSTM setup is optimized for handling and predicting from sequential data, making it particularly effective in applications involving time series analysis or any context where data is temporally correlated. The model's architecture, with its two LSTM layers and dense output layers, allows for accurate prediction of time-dependent patterns while minimizing errors through efficient optimization.

5) Simple RNN Model: One type of neural network that is arranged collectively, with the primary purpose of enabling each network to interact with another network, is the simple RNN [35]. The model includes a single simple RNN layer with 100 units using ReLU activation, aimed at capturing temporal data dependencies. It also incorporates a dropout layer with a rate of 0.2 to prevent overfitting. A single-unit dense layer follows, designed for producing continuous regression outputs. The model is trained for 100 epochs with a batch size of 64, which balances computational efficiency and model effectiveness. Training employs the Adam optimizer, with a lower learning rate, and the MSE loss function, facilitating efficient learning and accurate error measurement. This tuned simple RNN setup is optimized for straightforward yet effective time-series analysis, ensuring reliable predictions with minimal complexity. It leverages enhanced units and dropout regularization to improve performance while maintaining simplicity

6) Deep RNN Model: The architecture consists of three sequential simple RNN layers, each with 50 units. The ReLU activation function is used across all layers. The first two layers are configured to return sequences, preserving temporal information for deeper processing, while the final layer outputs a flattened vector. A single-unit dense layer follows the RNN layers, designed specifically for regression outputs. The model is trained over 100 epochs with a batch size of 64, providing a balance between computational efficiency and model performance. The Adam optimizer is employed for its efficiency in converging to optimal solutions, and the MSE loss function is used to quantify the accuracy of the model's predictions. This multilayer RNN setup is engineered to enhance the model's capability in analyzing and predicting complex, sequential data, making it well-suited for regression analysis in scenarios where time-dependent data is critical. The deeper architecture allows the model to effectively handle and predict from sequential data, ensuring accurate regression results.

7) Gated Recurrent Unit AE: The GRU AE employs gated GRU layers for both encoding and decoding in an AE framework. The encoder part comprises a bidirectional GRU layer with 64 units followed by a single GRU layer with 32 units, using ReLU activation to introduce nonlinearity. The encoded representation is then prepared for decoding using a RepeatVector layer, which repeats the context vector to match the input length. The decoder consists of two GRU layers, each with 32 units and ReLU activation, followed by a bidirectional GRU layer with 64 units to reconstruct the original input. Residual connections are added to improve reconstruction accuracy, while the final dense layer outputs a 2-D vector representing the denoised signals. The model is trained over 200 epochs with a batch size of 64, ensuring sufficient iterations for learning while maintaining stability and performance. The Adam optimizer is employed for adaptive learning rate management, and the MSE loss function measures reconstruction error, optimizing the model to produce outputs as close to the original inputs as possible. This enhanced GRU AE effectively handles sequential data, making it suitable for tasks such as time-series prediction, anomaly detection, and complex feature extraction from time-dependent sources.

8) Traditional Denoising Methods: The parameters are optimized as follows. For the *Wavelet Denoising*, the wavelet type used is "db4" (Daubechies wavelet of order 4), with a level of wavelet decomposition set to 3, meaning the signal is decomposed into three levels of detail coefficients. The thresholding method applied during denoising is "soft," with a threshold value determined by 0.25 times the median absolute deviation of the detail coefficients, scaled for noise estimation. For phase denoising, the parameters differ slightly, using a "coif1" wavelet type with "hard" thresholding, and a decomposition level of 4. The denoised signals are then rescaled to their original scale, with optional Gaussian smoothing applied for further refinement. This approach ensures efficient denoising of both magnitude and phase data, preserving the underlying signal structure. For the *Savitzky–Golay filter*, the window length was set to 21 and polynomial order was set to 3. The *moving average filter* is applied with a window size of 50, which smooths the data by averaging it over a specified window. A larger window size can potentially smooth out higher frequencies of the signal, which may distort the signal's frequency content. However, if the frequency of the signal is lower than the filter range, no significant changes in frequency are expected. The *low-pass filter* is designed with an order of 4, applying a Butterworth filter to ensure smooth frequency response. The filter uses a soft thresholding approach with a threshold multiplier of 0.1 to effectively reduce high-frequency noise while retaining the essential components of the signal. The cutoff frequency is set to 0.05 (normalized), allowing for the attenuation of higher frequencies while preserving the lower frequency. *Kalman filter* had 0 as an initial state mean and the number of dimensions observed was 1. The window size in *median filtering* was set to 21. *TV Denoising's* weight was 0.8 and *Wiener Filter's* "mysize" parameter was set to 15. The kernel in the *Gaussian processes* was 1.0 * radial basis function (length_scale = 1.0) + white

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF THE APPROACHES TAKEN

Model	Denoising Eff.	Comput. Complex.	Adaptability to Noise	Impl. Complex.
CNN	High	High	Requires extensive training	Complex
Random Forest Regressor	High	Moderate	Moderate	Moderate
Autoencoders	High	High	Requires extensive training	Complex
GRU Autoencoders	High	High	Requires extensive training	Complex
Simple RNN	High	High	Requires extensive training	Moderate
Deep RNN	High	High	Requires extensive training	Moderate
LSTM	High	High	Requires extensive training	Moderate
Wavelet Denoising	High	Moderate	Adaptable to various microresonator noise types	Moderate
Savitzky-Golay Filter	Moderate	Low	Effective for removing noise from microresonator signals	Simple
Moving Average	Moderate	Low	Suitable for smoothing microresonator signals	Simple
Low-pass filter	Moderate	Low	Suitable for filtering high-frequency noise from microresonator signals	Simple
Kalman Filter	High	Moderate	Effective for dynamic microresonator signals with noise	Moderate
Median Filtering	Moderate	Low	Effective for removing impulsive noise from microresonator signals	Simple
Wiener Filter	High	Moderate	Suitable for stationary microresonator signals	Moderate
Total Variation Denoising	High	Moderate	Effective for preserving microresonator signals features	Moderate
Gaussian Processes	High	High	Adaptable for various noise distributions in microresonator signals	Moderate

kernel (noise_level $= 0.1$) with the number of restarts for the optimizer of 10.

These models are meticulously designed and trained to effectively denoise microresonator signals, with each architecture specifically tuned to capitalize on particular features of the signal data and uniquely reduce noise. Table I illustrates a comparison of the various approaches employed.

Machine learning is a key component of our strategy for several important reasons.

1) Electronic noise in microresonators can be intricate and nonlinear, posing challenges to conventional noise reduction techniques. Machine learning has the ability to discern complex patterns within the data, proving to be an efficient solution to these issues [34].
2) Microresonators must cope with numerous noise sources that vary based on the specific application and environmental conditions. Machine learning algorithms are well-suited to adapt to these variations, ensuring effective noise reduction across diverse settings [35].
3) Machine learning's reliance on data enables the identification of subtle signal features and noise traits that traditional rule-based methods may miss. This approach allows for tailored noise reduction, thereby improving the quality of microresonator measurements [36].
4) Machine learning models continually refine and enhance their performance as they process more data. This capability is particularly valuable in long-term microresonator applications, where noise properties may evolve over time [37].

III. RESULTS AND ANALYSIS

A. Dataset Description

The dataset that was used as training data for the models includes 100 S_{21} signals (pixels) from microresonators, with each signal consisting of 1001 samples that represent frequency-amplitude pairs. These signals are exposed to a range of noise levels commonly encountered in experimental settings, necessitating effective denoising for precise analysis.

B. Assessment Measures

The quality factor, commonly referred to as Q, serves as a crucial metric in evaluating the most effective model for denoising signals from microresonators. It is defined as the ratio of a resonant system's peak or resonance frequency to its bandwidth.

The formula for the Q-factor is given as follows:

$$Q = \frac{f_{\max}}{\Delta f}.$$

To assess the changes in the quality factor of the signal following denoising, we undertake the following steps:

1) Calculate the quality factor (Q) of the original noisy signals using a predetermined metric, establishing a baseline for initial signal quality.
2) Implement denoising methods on the noisy microresonator signals to produce denoised outputs.
3) Recompute the quality factor (Q) for the denoised signals employing the same metric as used initially.
4) Evaluate the change in the quality factor postdenoising to determine the efficacy of the denoising techniques in improving the signal quality of the microresonators.

To determine if the denoising models have effectively enhanced the quality of the microresonators' signals, the following methods can be considered:

1) Peak frequency preservation: We ensure that the principal frequencies of the microresonator signals are maintained postdenoising. Any alterations in these frequencies might suggest that the denoising process introduced distortions.
2) Peak amplitude preservation: It is important to verify that the peak amplitudes of the microresonator signals remain

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

TABLE II
EXPERIMENTAL RESULTS AFTER TRAINING MODELS (MAGNITUDE)

Model	Q factor (Initial)	Q factor	Δf (Hz)	ΔA (dB)	MSE (dB)	SNR (dB)
Convolutional Neural Networks	15821.28	16061.29	1428.38	1.089	0.224	28.06
Deep Recurrent Neural Networks	15821.28	16063.84	70.06	0.328	0.426	25.26
Gaussian Processes	15821.28	15799.74	144.56	1.117	0.222	28.08
Gated Recurrent Units Autoencoders	15821.28	16265.90	1971.93	1.103	0.265	27.32
Kalman Filter	15821.28	15710.43	149.64	0.645	0.090	32.01
Long-Short Term Memory	15821.28	15954.56	574.50	0.392	0.170	30.12
Moving Average	15821.28	15824.67	153.56	0.420	0.115	31.46
Savitzky-Golay Filter	15821.28	15888.77	203.41	0.783	0.198	27.94
Total Variation Denoising	15821.28	15758.84	67.31	1.055	0.244	26.92
Wavelet Denoising	15821.28	15945.34	163.09	0.990	0.201	29.10
Wiener Filter	15821.28	15883.40	18.45	0.254	0.069	34.77

consistent after denoising. Shifts in this area could be indicative of distortion due to the denoising process.

3) Statistical analysis: Statistical tools, such as MSE, are utilized to quantitatively assess and compare the denoised signals to their original noisy counterparts.
4) SNR: Calculating the SNR of the denoised signals compared to the original noisy signals provides a quantitative measure of the improvement in signal quality. Higher SNR values indicate better denoising performance, as the signal becomes more prominent relative to the noise. The formula for the SNR is given as follows:

$$\mathrm{SNR} = 10 * \log_{10} \frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}.$$

The denoised signal close to the ideal (with minimum change in frequency and amplitude, minimum MSE, higher SNR, and a positive change in quality factor) would be a signal represented by Fig. 4(a).

TABLE III
EXPERIMENTAL RESULTS AFTER TRAINING MODELS (PHASE)

Model	MSE (Degrees)	SNR (dB)
Convolutional Neural Networks	10.483	31.79
Deep Recurrent Neural Networks	18.471	29.33
Gaussian Processes	10.343	31.84
Gated Recurrent Units Autoencoders	13.638	30.64
Kalman Filter	4.424	35.53
Long-Short Term Memory	7.891	33.12
Moving Average	5.486	34.28
Savitzky-Golay Filter	11.238	31.10
Total Variation Denoising	12.301	30.25
Wavelet Denoising	9.457	32.02
Wiener Filter	5.987	34.01

C. Results

The comprehensive evaluation of each denoising technique's effectiveness highlights the best method for improving the reliability of microresonators.

Figs. 4(b) and 5(b) present an initial original noisy signal's magnitude and phase, respectively, while Figs. 4(c)–(r) and 5(c)–(r) display the corresponding denoised signals produced by various denoising models for magnitude and phase. These figures contrast the original noisy signals with their denoised counterparts, labeled as noisy-to-clean signals.

Furthermore, Table I displays comparative performance measures for the different models.

D. Discussion

Conventional signal processing techniques exhibit specific advantages in preserving both magnitude and phase features while effectively reducing noise. Upon reviewing Figs. 4 and 5, it becomes clear that among traditional denoising methods, Kalman filter, Wiener filter, and wavelet denoising emerge as the most effective for both magnitude and phase components of the signal. These models not only excel in maintaining amplitude and frequency integrity but also manage to minimize phase noise, thereby enhancing the overall clarity and fidelity of the signal representation.

From Tables II and III, it can be observed that Kalman filter and Wiener filter maintain relatively low changes in resonant frequency (e.g., $\Delta f = 149.64$ Hz and 18.45 Hz, respectively) and amplitude (e.g., $\Delta A = 0.645$ dB and 0.254 dB, respectively) while achieving a low MSE for magnitude denoising. Similarly, wavelet denoising achieves minimal phase distortion with an MSE of 9.457°, coupled with a high SNR of 32.02 dB, indicating its capability to maintain phase integrity while reducing noise interference.

Other signal processing methods, such as moving average, Savitzky–Golay filter, and TV Denoising, are also effective in reducing noise while preserving signal properties. These approaches demonstrate relatively stable resonant frequency shifts (e.g., $\Delta f = 153.56$ Hz, 203.41 Hz, and 67.31 Hz, respectively) and minimal amplitude changes (e.g., $\Delta A = 0.420$ dB, 0.783 dB,

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

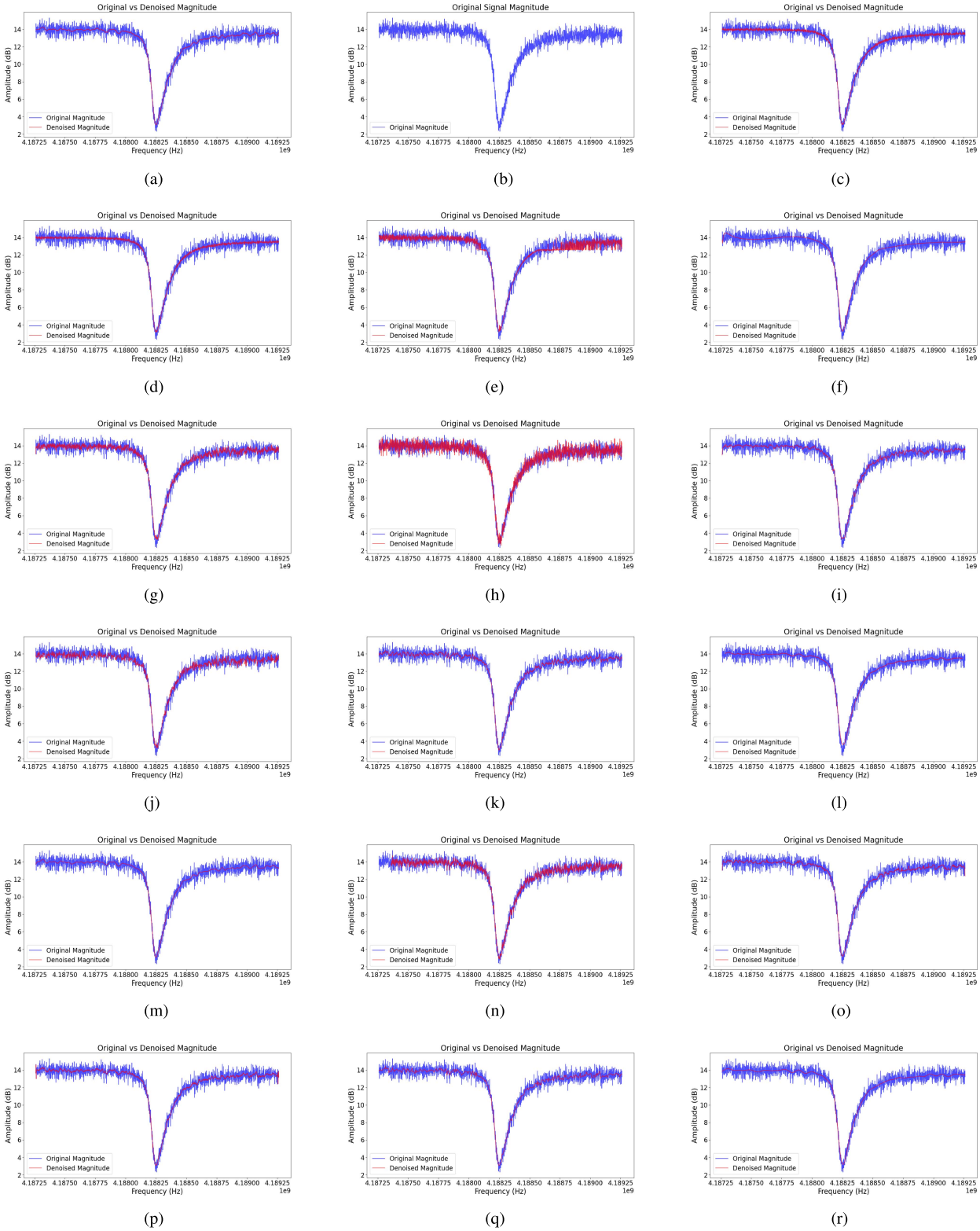

Fig. 4. Comparison of methods for noisy-to-clean signal's magnitude transformation. (a) Magnitude close to ideal output. (b) Initial noisy magnitude. (c) Noisy-to-clean magnitude using CNNs. (d) Noisy-to-clean magnitude using RFR. (e) Noisy-to-clean magnitude using sparse AE. (f) Noisy-to-clean magnitude using gated recurrent units (GRU) AE. (g) Noisy-to-clean magnitude using simple RNNs. (h) Noisy-to-clean magnitude using deep RNNs. (i) Noisy-to-clean magnitude using LSTM. (j) Noisy-to-clean magnitude using wavelet denoising. (k) Noisy-to-clean magnitude using Savitzky–Golay filter. (l) Noisy-to-clean magnitude using moving average. (m) Noisy-to-clean magnitude using low-pass filter. (n) Noisy-to-clean magnitude using Kalman filter. (o) Noisy-to-clean magnitude using median filtering. (p) Noisy-to-clean magnitude using Wiener filter. (q) Noisy-to-clean magnitude using TV denoising. (r) Noisy-to-clean magnitude using Gaussian processes.

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

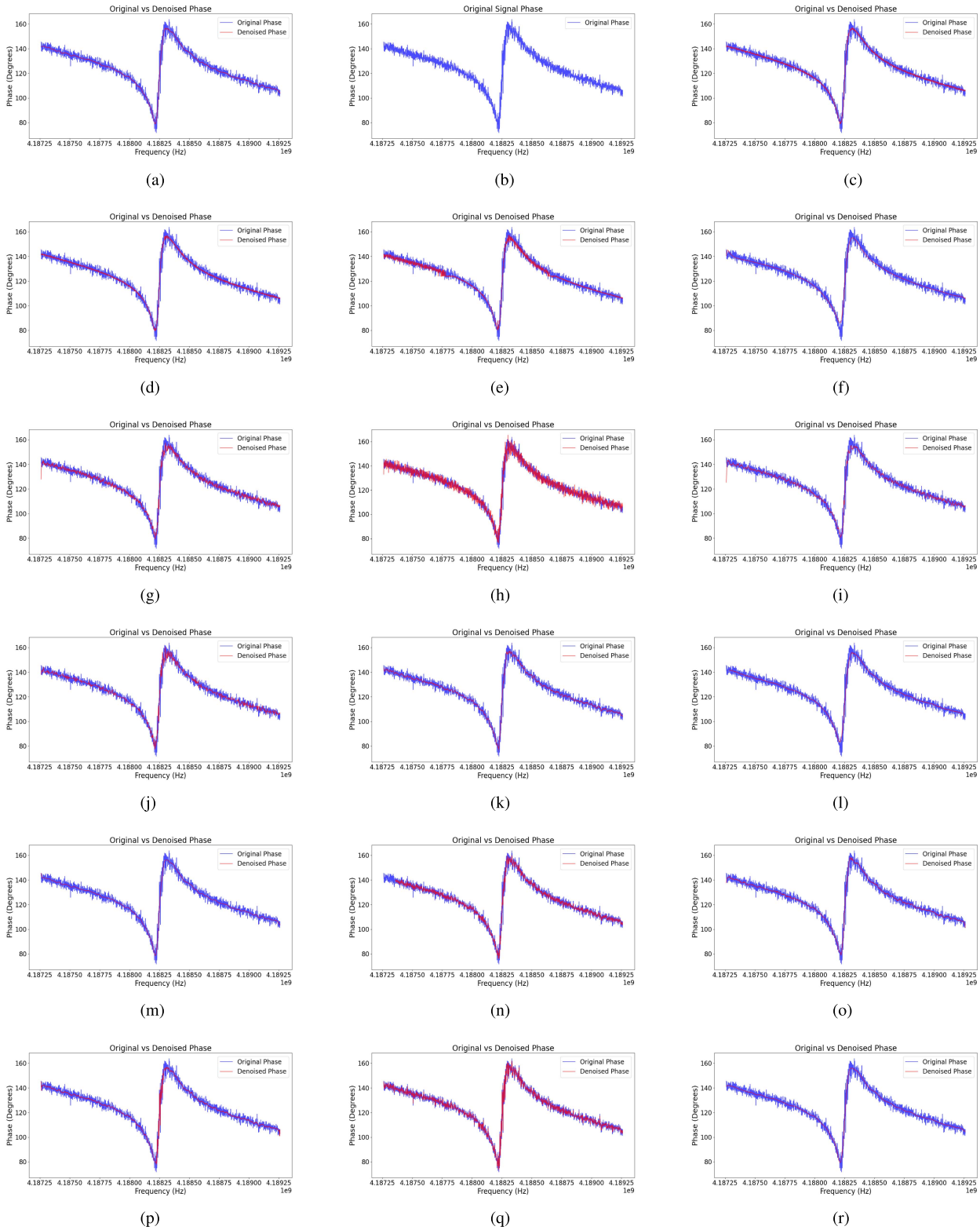

Fig. 5. Comparison of methods for noisy-to-clean signal's phase transformation. (a) Phase close to ideal output. (b) Initial noisy phase. (c) Noisy-to-clean phase using CNNs. (d) Noisy-to-clean phase using RFR. (e) Noisy-to-clean phase using sparse AE. (f) Noisy-to-clean phase using GRU AE. (g) Noisy-to-clean phase using simple RNNs. (h) Noisy-to-clean phase using deep RNNs. (i) Noisy-to-clean phase using LSTM. (j) Noisy-to-clean phase using wavelet denoising. (k) Noisy-to-clean phase using Savitzky–Golay filter. (l) Noisy-to-clean phase using moving average. (m) Noisy-to-clean phase using low-pass filter. (n) Noisy-to-clean phase using Kalman filter. (o) Noisy-to-clean phase using median filtering. (p) Noisy-to-clean phase using Wiener filter. (q) Noisy-to-clean phase using TV denoising. (r) Noisy-to-clean phase using Gaussian processes.

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

and 1.055 dB, respectively). Although these methods achieve a smaller MSE for magnitude signals, they are slightly less effective in phase denoising, with SNRs lower than those of Kalman filter and Wiener filter.

Overall, a clear pattern emerges from these results: when the MSE is low, the SNR is high, and vice versa, for both magnitude and phase denoising. This correlation could be attributed to the intrinsic properties of these traditional approaches, which filter noise within specific frequency ranges, potentially altering the frequency composition of the signals. However, models such as Kalman filter and Wiener filter exhibit better adaptability, preserving the signal's core features across both magnitude and phase components.

In addition, we used an established method known as the Noise2Noise method to train deep learning models such as CNNs, simple RNNs, deep RNNs, and RFRs without a clean pulse underlying as a true basis [27]. The Noise2Noise process is a method for training a denoising model without using clean target data [38]. The Noise2Noise model appears to eliminate the majority of the noise without causing significant distortion. Although it did not outperform any of the conventional samples used to train simulated pulses, the most important limitation of the Noise2Noise method, which outperformed the traditional denoising method in terms of quality factor, was the amount of data and quality of data to train it. More data, especially from different sources and larger energy ranges, will allow the network to see more variations in the pulses, thus improving the model. Because of taking Noise2Noise method (considering its limitations) with these models and having no pretrained or labeled clean data, we can see the difference in the denoising efficacy of the models from Table I. If we had some clean data to train our machine learning models on, we could have a high denoising efficacy as stated in the Table I.

CNN, RFR, simple RNN, and deep RNN demonstrated varied resonant frequency changes, with the deep RNN achieving a minimal shift of 70.06 Hz, while CNNs showed a larger shift of 1428.38 Hz, indicating a greater frequency alteration in magnitude denoising. Despite these frequency variations, amplitude changes remained moderate across these models, and the MSE was significantly reduced. This suggests that while each model's denoising effectiveness differs slightly, the overall trend of modest amplitude change and low MSE indicates a favorable outcome for magnitude denoising.

Moreover, Tables II and III reveal that some machine learning models performed exceptionally well in denoising both magnitude and phase components of microresonator signals. Models such as GRU AEs and LSTM networks were particularly effective in improving signal quality while maintaining a low MSE. However, their impact on signal properties varied: GRU AEs managed to maintain a stable core frequency shift ($\Delta f = 0$ Hz) but introduced amplitude changes, achieving a relatively low MSE in magnitude denoising. In contrast, LSTM models showed minor amplitude variations but exhibited more noticeable frequency shifts.

In terms of phase denoising, GRU AEs and wavelet denoising achieved good results, with low phase MSEs of 13.638° and 9.457°, respectively, while maintaining comparatively high SNRs. Notably, Kalman filter demonstrated the best performance in phase denoising, achieving the lowest phase MSE (4.424°) and the highest SNR among all models tested. However, the GRU AE, while achieving a high SNR for magnitude denoising (28.82 dB), had relatively lower SNR for phase denoising. The observed differences in the behavior of these models can be attributed to their intrinsic architectures and training procedures. For instance, AEs tend to prioritize reconstruction fidelity, which can result in amplitude changes to better represent the input signals, especially in magnitude denoising. In contrast, LSTM models, which are designed to capture temporal dependencies, tend to focus more on retaining frequency dynamics, resulting in more significant frequency variations but less impact on amplitude. Similarly, in phase denoising, models such as GRU AEs aim to maintain phase stability while minimizing MSE, which aligns with their architecture's focus on sequence modeling.

The Kalman filter was expected to have a high denoising efficiency. However, it only performed somewhat well in terms of SNR, magnitude and phase data. The Kalman filter's reliance on dynamic signal assumptions, which may not be consistent with our microresonator signal's specific noise structure, could be the source of this difference.

TV denoising, which was intended to give outstanding denoising efficiency, actually performed relatively well; SNR values demonstrated little noise reduction in both phase and magnitude data. This gap is most likely due to TV denoising's retention of extra signal features, which may limit its ability to totally suppress sophisticated noise patterns in this particular application while maintaining signal integrity.

Because Gaussian processes can adapt to various noise topologies, they were expected to have high denoising efficiency. Despite their great magnitude denoising performance, they had very low phase denoising efficiency, indicating a partial mismatch. The Gaussian processes' assumptions about noise distributions may not adequately account for the complexity of phase noise in microresonators, which could explain the decreased efficiency.

The sparse AE, which was predicted to have good denoising efficiency, only performed moderately, particularly in phase data, where its SNR was lower than expected. This gap may be due to the sparse AE's inability to handle complex noise structures, necessitating additional regularization methods or specialist tweaking to improve performance.

Despite having a high anticipated denoising efficiency, the deep RNN achieved only a low SNR in magnitude and phase denoising. This mismatch could be attributed to the model's high processing requirements and hyperparameter sensitivity, which made it difficult to optimize for microresonator's peculiar noise characteristics.

The limitations of traditional denoising methods are not only due to their slow response times with large datasets, which making them inefficient for fast and time-sensitive measurements, but also their reduced effectiveness in handling complex noise patterns and adapting to varying noise configurations. This highlights the need for machine learning methods. Traditional magnitude and phase denoising methods, such as the Wiener filter, wavelet denoising, and Kalman filter, offer low MSEs with

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

high SNRs, but their adaptability is limited by their assumptions about noise models. Despite the various noise patterns present in microresonator signals, the Kalman filter assumes that dynamic signal noise is constant.

In contrast, machine learning approaches offer a flexible, data-driven solution by directly learning noise patterns from the data. GRU AEs and LSTMs performed similarly in phase denoising, achieving a steady frequency shift and a magnitude denoising SNR of 28.82 dB, respectively. Furthermore, when compared to standard approaches, these models adapt to noise fluctuations more effectively. Despite training data quality constraints, such as the absence of clean signals, machine learning models such as CNNs and RNNs showed significant noise reduction, as seen by reduced MSE and minimal amplitude shifts. These findings indicate that machine learning models can outperform traditional approaches when complicated noise patterns or changing signal dynamics are inevitable, as long as there is a sufficient amount of training data.

Overall, the performance of each model in denoising both magnitude and phase components of microresonator signals is shaped by a complex interaction of factors, including network architecture, training data quality, and signal properties. Further research and experimentation are necessary to refine these models and determine the most effective applications of machine learning in this domain.

IV. Conclusion

In summary, this study provided an extensive evaluation of various denoising techniques for microresonator data, focusing on both magnitude and phase components. It encompassed classical signal processing methods alongside machine learning algorithms. This comprehensive analysis revealed insights into the strengths and limitations of each approach in modifying signal attributes such as the quality factor, MSE, SNR, frequency, and amplitude.

Traditional signal processing methods such as Kalman filter, Wiener filter, and wavelet denoising proved highly effective in reducing noise while preserving both magnitude and phase features, often achieving high SNRs. These methods were particularly adept at minimizing signal interference without causing significant changes to resonant frequency, amplitude, or phase, leading to a notable decrease in MSE.

Conversely, machine learning models such as CNNs, simple RNNs, deep RNNs, and RFRs faced challenges in denoising both magnitude and phase signals, primarily due to limitations in training data quality. These models struggled to generalize effectively with sparse or noisy data, resulting in difficulties with overfitting and inconsistent frequency preservation. However, models such as AEs and LSTM networks showcased promising results, particularly in phase denoising, with lower phase MSEs and better signal quality retention. The results of this study show the need for machine learning methods for denoising situations with intricate or variable noise patterns. Because ML models are data-driven learners, they can cope with a wider range of noise patterns than traditional techniques. The phase-denoising performance of LSTMs and GRU AEs demonstrates their ability to outperform traditional methods. Nonetheless, the availability of clean and diverse training data restricts their efficacy and flexibility, emphasizing the need to improve data collection and model training methods.

Finally, the efficacy of denoising microresonator signals hinges on a multifaceted interplay of factors including the chosen denoising approach, model architecture, availability of training data, and signal characteristics. Further research and experimentation are warranted to refine denoising methodologies and advance the application of machine learning in this domain. For instance, future endeavors could involve modeling pristine microresonator signals and broadening data collection efforts to empower supervised models for more effective learning.

References

[1] L. Fanals, E. Alarcón, I. Bashir, E. Blokhina, D. Leipold, and R. B. Staszewski, "Tunable *LC* resonator for multiplexed multi-qubit readout," in *Proc. IEEE Int. Symp. Circuits Syst.*, 2023, pp. 1–5.

[2] P. Szypryt, B. A. Mazin, B. Bumble, H. G. Leduc, and L. Baker, "Ultraviolet, optical, and near-IR microwave kinetic inductance detector materials developments," *IEEE Trans. Appl. Supercond.*, vol. 25, no. 3, Jun. 2015, Art. no. 2400604.

[3] S. McHugh et al., "A readout for large arrays of microwave kinetic inductance detectors," *Rev. Sci. Instrum.*, vol. 83, no. 4, Apr. 2012, Art. no. 044702, doi: 10.1063/1.3700812.

[4] Y. Sueno et al., "Characterization of two-level system noise for a microwave kinetic inductance detector comprising niobium film on a silicon substrate," *Prog. Theor. Exp, Phys.*, vol. 2022, no. 3, Feb. 2022, Art. no. 033H01, doi: 10.1093/ptep/ptac023.

[5] J. Li et al., "Quasiparticle generation-recombination noise in the limit of low detector volume," *IEEE Trans. Appl. Supercond.*, vol. 33, no. 5, Aug. 2023, Art. no. 2400105.

[6] J. v. Rantwijk, M. Grim, D. v. Loon, S. Yates, A. Baryshev, and J. Baselmans, "Multiplexed readout for 1000-pixel arrays of microwave kinetic inductance detectors," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 6, pp. 1876–1883, Jun. 2016.

[7] S. Probst, F. B. Song, P. A. Bushev, A. V. Ustinov, and M. Weides, "Efficient and robust analysis of complex scattering data under noise in microwave resonators," *Rev. Sci. Instrum.*, vol. 86, no. 2, Feb. 2015, Art. no. 024706, doi: 10.1063/1.4907935.

[8] B. A. Mazin, "Microwave kinetic inductance detectors: The first decade," *AIP Conf. Proc.*, vol. 1185, no. 1, pp. 135–142, Dec. 2009, doi: 10.1063/1.3292300.

[9] M. Shafiee, B. Grossan, J. Hu, I. Colantoni, and G. Smoot, "Design optimization of a 10 kilopixel optical band microwave kinetic inductance detector," *J. Instrum.*, vol. 14, no. 12, Dec. 2019, Art. no. P12011, doi: 10.1088/1748-0221/14/12/P12011.

[10] M. Shafiee, D. Fedorov, B. Grossan, V. Kizheppatt, and G. Smoot, "A readout system for microwave kinetic inductance detectors using software defined radios," *J. Instrum.*, vol. 16, no. 07, Jul. 2021, Art. no. P07015, doi: 10.1088/1748-0221/16/07/P07015.

[11] M. Shafiee, S. Feghhi, and J. Rahighi, "Experimental performance evaluation of ILSF BPM data acquisition system," *Measurement*, vol. 100, pp. 205–212, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0263224117300039

[12] C. P. Dautov and M. S. Ozerdem, "Wavelet transform and signal denoising using wavelet method," in *Proc. 26th Signal Process. Commun. Appl. Conf.*, 2018, pp. 1–4.

[13] N. S. Haider, R. Periyasamy, D. Joshi, and B. K. Singh, "Savitzky-Golay filter for denoising lung sound," *Braz. Arch. Biol. Technol.*, vol. 61, 2018, Art. no. e18180203, doi: 10.1590/1678-4324-2018180203.

[14] A. Tangborn and S. Q. Zhang, "Wavelet transform adapted to an approximate Kalman filter system," *Appl. Numer. Math.*, vol. 33, pp. 307–316, 2000.

[15] R. Pecen, S. Ula, and M. Timmerman, "Modeling and simulation of a Kalman filter based control scheme for an ac/dc power system," *Elect. Power Energy Syst.*, vol. 26, pp. 173–189, 2004.

[16] H.-C. Huang and T. Lee, "Data adaptive median filters for signal and image denoising using a generalized sure criterion," *IEEE Signal Process. Lett.*, vol. 13, no. 9, pp. 561–564, Sep. 2006.

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.

[17] S. S. Haykin, *Adaptive Filter Theory*. Englewood Cliffs, NJ, USA: Prentice Hall, 1996.

[18] S. Orfanidis, *Optimum Signal Processing*. Piscataway, NJ, USA: Rutgers Univ., 2007.

[19] B. R. Manju and M. R. Sneha, "ECG denoising using Wiener filter and Kalman filter," *Procedia Comput. Sci.*, vol. 171, pp. 273–281, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050920309947

[20] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," in *Proc. Int. Symp. Phys. Des.*, 1992, pp. 259–268.

[21] R. Ramakrishna, H.-T. Wai, and A. Scaglione, "A user guide to low-pass graph signal processing and its applications: Tools and applications," *IEEE Signal Process. Mag.*, vol. 37, no. 6, pp. 74–85, Nov. 2020.

[22] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2006.

[23] O. S. Singh and R. K. Sunkaria, "ECG signal denoising based on empirical mode decomposition and moving average filter," in *Proc. IEEE Int. Conf. Signal Process., Comput. Control*, 2013, pp. 1–6.

[24] G. Yao, T. Lei, and J. Zhong, "A review of convolutional-neural-network-based action recognition," *Pattern Recognit. Lett.*, vol. 118, pp. 14–22, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167865518302058

[25] A. Dhillon and G. Verma, "Convolutional neural network: A review of models, methodologies and applications to object detection," *Prog. Artif. Intell.*, vol. 9, pp. 85–112, 2020, doi: 10.1007/s13748-019-00203-0.

[26] G. Fan, J. Li, and H. Hao, "Vibration signal denoising for structural health monitoring by residual convolutional neural networks," *Measurement*, vol. 157, 2020, Art. no. 107651. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0263224120301895

[27] M. R. Anderson et al., "Performance of a convolutional autoencoder designed to remove electronic noise from p-type point contact germanium detector signals," *Eur. Phys. J. C*, vol. 82, no. 12, Dec. 2022, Art. no. 1084.

[28] F. Gao, B. Li, L. Chen, X. Wei, Z. Shang, and C. He, "Ultrasonic signal denoising based on autoencoder," *Rev. Sci. Instrum.*, vol. 91, no. 4, 2020, Art. no. 045104, doi: 10.1063/1.5136269.

[29] J. H. Graw, W. T. Wood, and B. J. Phrampus, "Predicting global marine sediment density using the random forest regressor machine learning algorithm," *J. Geophys. Res.: Solid Earth*, vol. 126, no. 1, 2021, Art. no. e2020JB020135. [Online]. Available: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2020JB020135

[30] Z. C. Lipton, J. Berkowitz, and C. Elkan, "A critical review of recurrent neural networks for sequence learning," 2015, *arXiv.1506.00019*.

[31] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[32] J. Schmidt-Hieber, "Nonparametric regression using deep neural networks with ReLU activation function," *Ann. Statist.*, vol. 48, no. 4, pp. 1875–1897, 2020, doi: 10.1214/19-AOS1875.

[33] Z. Zhang, "Improved Adam optimizer for deep neural networks," in *Proc. IEEE/ACM 26th Int. Symp. Qual. Serv.*, 2018, pp. 1–2.

[34] C. M. Bishop, *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Berlin, Germany: Springer-Verlag, 2006.

[35] A. Yapici and M. A. Akcayol, "A review of image denoising with deep learning," in *Proc. 2nd Int. Informat. Softw. Eng. Conf.*, 2021, pp. 1–6.

[36] Pawan and R. Dhiman, "Machine learning techniques for electroencephalogram based brain-computer interface: A systematic literature review," *Meas.: Sensors*, vol. 28, 2023, Art. no. 100823. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2665917423001599

[37] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, and S. Wermter, "Continual lifelong learning with neural networks: A review," *Neural Netw.*, vol. 113, pp. 54–71, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608019300231

[38] J. Lehtinen et al., "Noise2Noise: Learning image restoration without clean data," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, vol. 80, pp. 2965–2974.

Authorized licensed use limited to: Nazarbayev University. Downloaded on April 15,2026 at 15:16:12 UTC from IEEE Xplore. Restrictions apply.